CBSE 2027 Maths — Predicted Paper (Set 9 of 10)

Based on the highly weighted CBSE syllabus chapters and a rigorous analysis of latest competency-based trends from the official blueprint, here is Set 9 of the high-probability predicted Class 12 Maths board paper.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3x - 4$. What is the inverse function $f^{-1}(x)$? * **Chapter:** Relations and Functions * **Probability Tag: Very High**

Q2. Find the domain of the inverse trigonometric function $y = \sin^{-1}(x^2 - 4)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: High**

Q3. If A is a 3×3 invertible matrix such that $|A| = 4$, find the value of $|A^{-1}|$. * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q4. What is the maximum possible value of a 2×2 determinant whose elements are exclusively chosen from the set $\{0, 1\}$? * **Chapter:** Determinants * **Probability Tag: Medium**

Q5. What is the value of the derivative of x^x with respect to x at $x = e$? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The function $y = x - \sin x$ is strictly increasing for which values of x? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate the indefinite integral: $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} \, dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q8. Find the area bounded by the curves $y = \sin x$ and $y = \cos x$ between the ordinates $x = 0$ and $x = \frac{\pi}{4}$. * **Chapter:** Application of Integrals * **Probability Tag: Medium**

Q9. Find the integrating factor (I.F.) of the linear differential equation: $(1 - x^2)\frac{dy}{dx} - xy = 1$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q10. For what value of scalar p are the vectors $\vec{a} = 2\hat{i} + 3\hat{j} + p\hat{k}$ and $\vec{b} = p\hat{i} - \hat{j} + \hat{k}$ orthogonal (perpendicular)? * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. If for two non-zero vectors $\vec{a}$ and $\vec{b}$, the relation $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$ holds true, what is the angle between $\vec{a}$ and $\vec{b}$? * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. If a straight line has direction ratios proportional to a, a, a (where $a \neq 0$), determine its exact direction cosines. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q13. Find the coordinates of the point where the line $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+1}{4}$ intersects the yz-plane. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. In a Linear Programming Problem (LPP), if the objective function $Z = cx + dy$ reaches its maximum value at two distinct adjacent corner points of the feasible region, how many optimal solutions does the LPP have in total? * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. A fair coin is tossed 3 times. What is the conditional probability of getting exactly 2 heads, given that at least one head has occurred? * **Chapter:** Probability * **Probability Tag: High**

Q16. Evaluate the definite integral using standard limits properties: $\int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}}\,dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. State the order of the differential equation that represents the family of all concentric circles centered at the origin. * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. Two cards are drawn successively without replacement from a well-shuffled pack of 52 playing cards. Find the probability that both cards drawn are Kings. * **Chapter:** Probability * **Probability Tag: High**

Q19. (Assertion-Reason): Assertion (A): The determinant of a skew-symmetric matrix of an even order is always a perfect square. **Reason (R):** The determinant of any skew-symmetric matrix of an odd order is always equal to zero. * **Chapter:** Matrices and Determinants * **Probability Tag: Medium**

Q20. (Assertion-Reason): Assertion (A): The function $f(x) = |x|$ is not differentiable at $x = 0$. **Reason (R):** Every function that is continuous at a given point is necessarily differentiable at that point. * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Evaluate the principal value of $\sin(\tan^{-1} x + \cot^{-1} x)$, where $x \in \mathbb{R}$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. Find $\frac{dy}{dx}$ if $x^y = y^x$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q23. Find the area of a parallelogram whose diagonals are given by the vectors $\vec{d_1} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{d_2} = \hat{i} - 3\hat{j} + 4\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q24. Solve the differential equation by substitution: $\frac{dy}{dx} = \sin(x+y)$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. A pair of fair dice is thrown once. Find the probability of getting a total sum of 8 or 9. * **Chapter:** Probability * **Probability Tag: Medium**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Prove that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x}{x^2+1}$ is neither one-one nor onto. * **Chapter:** Relations and Functions * **Probability Tag: High**

Q27. If $A = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$, compute $(AB)^{-1}$ and verify that $(AB)^{-1} = B^{-1}A^{-1}$. * **Chapter:** Matrices * **Probability Tag: High**

Q28. Differentiate the function $y = (\sin x)^x + \sin^{-1} \sqrt{x}$ with respect to x. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate the indefinite integral using the half-angle trigonometric substitution method: $\int \frac{dx}{5+4\cos x}$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the linear differential equation: $(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. A man is known to speak the truth 3 out of 4 times. He throws a standard die and reports to his friend that it is a 6. Using Bayes' Theorem, find the exact probability that it is *actually* a 6. * **Chapter:** Probability * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the following definite integral using properties of limits: $\int_0^{\pi} \log(1+\cos x)\, dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. Show that the volume of the greatest right circular cylinder which can be inscribed in a given cone of height h and semi-vertical angle α is $\frac{4}{27}\pi h^3 \tan^2 \alpha$. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Using the method of integration, find the area of the region bounded by the parabola $y^2 = 4x$ and the circle $4x^2 + 4y^2 = 9$. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q35. Find the coordinates of the foot of the perpendicular drawn from the point $A(1, 8, 4)$ to the line joining the points $B(0, -1, 3)$ and $C(2, -3, -1)$. Hence, find the image of point A in the given line. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Automobile Production Costs) A car manufacturing company produces three different models: SUV, Sedan, and Hatchback. For each unit produced, the cost of materials, labour, and overheads are evaluated. The costs (in thousands of rupees) are: SUV: Materials 400, Labour 100, Overheads 50. Sedan: Materials 300, Labour 80, Overheads 40. Hatchback: Materials 200, Labour 60, Overheads 30. During a specific month, the factory produced 50 SUVs, 100 Sedans, and 150 Hatchbacks. *(i)* Express the cost of production for each vehicle model as a 3×3 matrix C and the monthly production output as a column matrix P. *(1 Mark) (ii)* Using matrix multiplication, find the total cost of materials spent by the company in that month. *(1 Mark) (iii)* Using matrix operations, calculate the total combined cost of production (materials + labour + overheads) for all models produced that month. *(2 Marks)* * **Chapter:** Matrices * **Probability Tag: High**

Q37. Case Study 2: Application of Derivatives (Fluid Dynamics) An industrial water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is being poured into the tank at a constant rate of $5 \text{ cubic metres per hour}$. Let h be the depth of water in the tank at any time t. *(i)* Express the volume V of the water in the tank purely in terms of the water depth h. *(1 Mark) (ii)* Find the exact rate at which the water level h is rising when the depth is 4 metres. *(1 Mark) (iii)* Find the rate of change of the wet surface area (curved surface area of the water) of the tank at the exact moment when the depth of the water is 4 metres. *(2 Marks)* * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Expected Values & Gambling) At a school carnival, a mathematics teacher devises a gambling game using a fair six-sided die. The rules are: a player wins ₹5 for throwing a number strictly greater than 4 (i.e., a 5 or 6), and loses ₹1 for throwing any other number. A student decides to play the game with a specific strategy: he will throw the die up to a maximum of three times, but he will immediately quit the game *as soon as* he gets a number greater than 4. Let X denote the total amount won or lost. *(i)* What is the probability that the student wins the game on his very first throw? *(1 Mark) (ii)* What is the probability that the student ends up throwing the die exactly two times before quitting? *(1 Mark) (iii)* Create the probability distribution for the total amount won/lost (X) and calculate the expected value (mean amount) the student wins or loses per game. *(2 Marks)* * **Chapter:** Probability * **Probability Tag: High**
